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REVIEW ARTICLE

Integrating Generative AI in Higher Art Education: A Systematic Review of Tools, Pedagogies, and Practices

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Abstract

The rapid expansion of Generative Artificial Intelligence (GenAI) is transforming higher art education, raising new questions about creativity, authorship, and skill development. This systematic review synthesizes 27 empirical and observational studies published between 2023 and 2025 to examine how GenAI tools—including ChatGPT, Midjourney, and DALL·E—are integrated across visual arts, design, music, and interdisciplinary STEAM education. Following PRISMA guidelines, the reviewed studies were systematically screened, appraised, and thematically analyzed to identify pedagogical roles, learning outcomes, technical deployment modes, and ethical concerns. Across disciplines, GenAI consistently functions as a powerful support for early-stage creative exploration, enhancing divergent thinking and reducing affective barriers, particularly for novice learners. However, the synthesis identifies a persistent ideation–execution gap: while GenAI accelerates conceptual generation, it remains limited in supporting executional phases that require technical nuance, embodied practice, and contextual judgment. Discipline-specific patterns indicate stronger quantitative evidence of creative fluency gains in music education, alongside ongoing concerns about authenticity and de-skilling in visual arts and design contexts. This review advances the field by linking creative outcomes to both pedagogical framing and technical deployment modes, and argues for Human-in-the-Loop pedagogies that position students as critical curators. Embedding critical AI literacy, including prompt engineering, ethical awareness, and algorithmic understanding, is essential for the sustainable integration of GenAI in higher art education.

Keywords Art education · Creativity · Generative AI · Higher education · Instructional design · Pedagogy

Introduction

The year 2022 marked a pivotal turning point in the evolution of Generative Artificial Intelligence (GenAI), particularly with the public release of ChatGPT, which dramatically transformed how individuals engage with information, perform creative tasks, and interact within educational contexts [7]. This shift catalyzed a surge of interest in AI applications across disciplines, as generative models demonstrated remarkable capabilities in producing human-like outputs. Since then, GenAI has expanded into a multimodal ecosystem, encompassing various forms of content generation, including text (e.g., ChatGPT), images (e.g., Midjourney), music (e.g., MuseNet), and even video (e.g., Sora). These advancements have profound implications for the design, delivery, and assessment of learning, especially in creative fields [39].

In response to the evolving demands of the 21st-century workforce, the effective integration of GenAI into higher art education has been increasingly recognized as a pathway to cultivating critical competencies such as creativity, innovation, and divergent thinking [18]. Emerging empirical studies have begun to shed light on the nuanced ways GenAI influences learning outcomes in the arts. For example, Doshi and Hauser [14] conducted a causal study examining the impact of GenAI on creative writing, revealing that GenAI helped participants—especially those with lower prior ability—generate more compelling narratives. However, they also identified a homogenizing effect, where GenAI-assisted stories resembled one another more closely than those created solely by humans. Similarly, Hubert et al. (2024) found that GenAI, particularly ChatGPT, demonstrated superior

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fluency and idea generation in tasks requiring divergent thinking, highlighting its potential as a creativity-enhancing tool. These studies underscore GenAI's growing role in augmenting artistic ideation and production, contributing to an expanding body of literature exploring its application across various art domains, including music composition [12], visual arts [29], and film production [16].

Despite the promising potential of GenAI, its integration into art education presents a range of challenges. Scholars have raised concerns about copyright and intellectual property issues related to AI-generated content [40], as well as broader ethical concerns regarding transparency, authorship, and academic integrity [48]. Additionally, the widespread use of GenAI could inadvertently exacerbate issues of access, equity, and representation, particularly in institutions lacking the resources to support its use, or when dominant GenAI models marginalize underrepresented voices [22, 27]. These concerns underscore the need for a transformative approach to art education—one that embraces technological innovation while critically interrogating its broader implications [4, 5]. Scholars have proposed several pedagogical responses, such as developing AI-powered pedagogies [43], cultivating AI thinking [24], and applying prompt engineering strategies to better engage with AI tools [26]. Others advocate for critical pedagogies that empower students to question and analyze the biases embedded within AI systems [37] and promote AI literacy to ensure informed and ethical engagement with these technologies [15].

In light of recent technological and pedagogical advancements, this systematic review synthesizes 27 empirical studies conducted between 2024 and 2025 to explore how GenAI tools—such as ChatGPT, Midjourney, and DALL·E—are being integrated into higher art education across creative disciplines, including visual arts, animation, music, and dance. This review moves beyond providing a unified pedagogical model, as the studies reflect a diverse range of GenAI applications, including idea generation, visual ideation, feedback personalization, and multimodal interaction. Although most studies did not explicitly specify pedagogical frameworks, their instructional practices implicitly align with approaches such as project-based learning, studio-oriented creation, and technology-mediated critique. This review also explores the perspectives of both students and educators, highlighting the pedagogical affordances of GenAI—such as enhanced creativity, increased motivation, and personalized support—while addressing emerging concerns about academic integrity, ethical use, and the potential homogenization of creative expression. Faculty perspectives reveal a cautious optimism, tempered by challenges related to institutional readiness, access disparities, and the evolving role of the educator. By synthesizing these findings, this review offers a critical and practice-informed account of how GenAI can be meaningfully and responsibly integrated into the teaching and learning of art and design. Ultimately, it contributes to ongoing conversations about the future of higher art education by advocating for pedagogical innovation grounded in ethical awareness and inclusive design.

Method

This systematic review was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [38] to ensure transparency and methodological rigor in the identification, selection, analysis, and synthesis of literature examining pedagogical uses of generative artificial intelligence (GenAI) in higher art education.

Eligibility Criteria

The inclusion criteria were defined using the PICoS framework (Population, Interest, Context, and Study type) [6]. The population included students and instructors engaged in higher education art programs, encompassing visual arts, media arts, design, music, and interdisciplinary creative disciplines. The phenomenon of interest was the pedagogical application of GenAI tools, including but not limited to ChatGPT, Midjourney, DALL·E, and RunwayML, within teaching and learning activities. The context was restricted to formal higher education settings, such as universities, colleges, and art academies, at undergraduate and postgraduate levels. Eligible sources comprised peer-reviewed journal articles, conference proceedings, and empirical or observational studies published between January 2022 and July 2025. Only studies published in English were included.

Search Strategy

A comprehensive literature search was conducted across two major academic databases: Scopus and Web of Science. These databases were selected for their multidisciplinary coverage, rigorous indexing standards, and inclusion of high-impact journals relevant to generative AI, pedagogy, and art education. The search strategy combined controlled vocabulary and free-text keywords related to generative AI, art education, higher education, and pedagogy. Boolean operators were used to connect concept groups with AND, while related terms within each group were combined using OR. A sample Boolean search string is presented in Table 1.

Table 1 Concept and keywords used to identify relevant articles

Concept	Related keywords
GenAI	"generative AI" OR GenAI OR "AI-generated art" OR "text-to-image" OR ChatGPT OR Midjourney OR DALL·E
Art Education	"art education" OR "arts education" OR "creative education" OR "arts-based learning" OR "arts-based pedagogy" OR "music education" OR "visual art" OR "media art" OR "performing arts" OR "interdisciplinary arts"
Higher Education pedagogy	"higher education" OR university OR college teaching OR pedagogy OR "instructional strategies" OR curriculum

Screening & Selection

All retrieved records were imported into reference management software (Zotero), and duplicate entries were removed. Titles and abstracts were screened against the eligibility criteria. Full-text articles deemed potentially relevant were subsequently assessed for inclusion. The study selection process is summarized in a PRISMA flow diagram (Fig. 1), which documents the number of records identified, screened, excluded, and included at each stage (Table 2).

Data Extraction

Data were extracted using a structured extraction form to ensure consistency across studies. Extracted information included: author(s), year of publication, and country; educational context (discipline and institutional level); type of GenAI technology employed; mode of tool deployment (e.g., web-based interface platforms such as ChatGPT web or Midjourney via Discord, versus API-based or locally hosted implementations such as Stable Diffusion); pedagogical model or instructional strategy; study design and methodology; participant characteristics; and key findings related to learning outcomes, pedagogical affordances, limitations, and ethical considerations.

Quality Appraisal

The methodological quality of included studies was assessed using appraisal tools appropriate to their research design. Empirical studies were evaluated using the Mixed Methods Appraisal Tool (MMAT, 2018 version) [23], while conceptual or theoretical papers were assessed using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Text and Opinion Papers (Joanna Briggs Institute, 2017). Quality appraisal was used to inform interpretation of findings rather than as an exclusion criterion.

Thematic Synthesis and Analytical Framework

Following data extraction, a qualitative thematic synthesis was conducted to identify patterns across the included studies. First, each study was systematically coded according to predefined analytical dimensions, including GenAI tool type, pedagogical intent, stage of the creative process addressed, and reported educational outcomes. Second, codes were iteratively compared across studies to identify recurring relationships and contrasts. Third, related codes were grouped into higher-order themes through an inductive process, resulting in a set of cross-cutting thematic categories that characterize current pedagogical uses of GenAI in higher art education.

PRISMA Flow of Study Selection

Following PRISMA 2020 guidelines, a total of 64 records were initially identified through systematic database searches. After the removal of 13 duplicate records, 51 unique records remained for title and abstract screening. During this screening phase, 8 records were excluded because they were not published in English or did not address both higher education and art- or creativity-related educational contexts.

As a result, 43 full-text articles were assessed for eligibility. Of these, 16 studies were excluded at the full-text review stage due to one or more of the following reasons: (1) limited relevance to generative AI within art or creative education contexts; (2) absence of empirical, observational, or practice-based evidence; or (3) a primary focus on technical system development without pedagogical application.

Ultimately, 27 studies met all inclusion criteria and were retained for the final qualitative synthesis. A detailed overview of the study selection process and reasons for exclusion at each stage is presented in Fig. 1 (PRISMA flow diagram).

Thematic Synthesis and Analytical Procedure

Following study selection, a qualitative thematic synthesis was conducted to identify recurring patterns in how generative artificial intelligence (GenAI) is conceptualized and applied within higher art education. Given the methodological heterogeneity of the included studies—spanning experimental designs, case studies, surveys, and qualitative reflections—a narrative and inductive synthesis approach was adopted rather than a statistical meta-analysis.

The analysis proceeded in three iterative stages. First, each included study was systematically coded according to descriptive categories, including (1) creative discipline (e.g., visual arts, music, design), (2) type of GenAI tool used (e.g., text-to-image, text-based language models, music

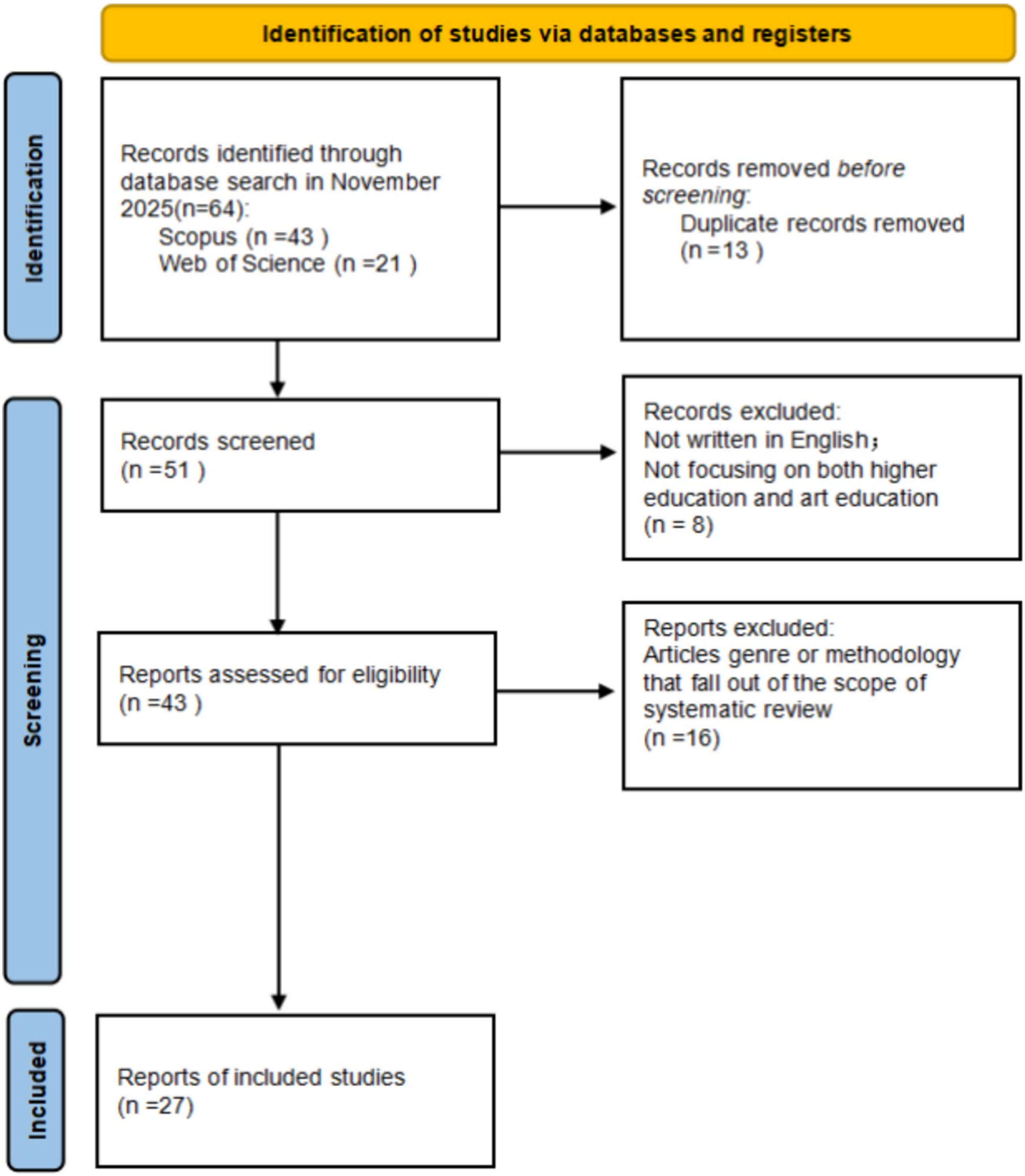

Fig. 1 PRISMA diagram of the study screening process and article selection

generation systems), (3) pedagogical function (e.g., ideation support, skill practice, feedback, reflection), and (4) reported learning outcomes and challenges. This initial coding focused on how GenAI was positioned within the teaching–learning process rather than on technical performance metrics.

Second, these descriptive codes were compared across studies to identify higher-order analytical themes related to pedagogical roles, learner experience, and instructional

Table 2 Comprehensive summary of reviewed studies (2023–2025)

Author (s) & Year	Context/discipline	Participants	GenAI Tool (s)	Methodology	Key focus/outcome
Ansone et al. [1]	Visual Arts (Lecture)	10 Art Students	DALL-E	Mixed-methods (Obs. + Quest.)	Ideation vs. Execution; Frustration with nuance
Khasawneh et al. [30]	Multimedia Design	60 Students	Midjourney	Quasi-experimental	Animation character design efficiency
Chen et al. [8]	Design History	64 Undergrads	ChatGPT	Quasi-experimental	Motivation, Academic Achievement, Self-Efficacy
Chiu and Hwang [11]	Art Appreciation	45 Students	Mind Mapping + GenAI	Experimental	Critical thinking awareness; Creative performance
Zhou and Kim (2024)[46]	Music Education	74 Undergrads	ChatGPT-4	Mixed-methods	Musical knowledge acquisition; Perception of ease
Chen et al. [9]	AI Art Edu. Alliance	108 Teachers	AI Tools (General)	Group Interviews + Quest	Institutional readiness; Art college caution vs. Uni readiness
Chen et al. [10]	Animation/Digital Media	108 Educators	AI Tools	Quest. + Interviews	Acceptance in animation; Creativity vs. over-reliance
Ma et al. [36]	Dance Education	4 Educators	ChatGPT	Bi-ethnographic	Human–machine interaction patterns in dance pedagogy
Wang [42]	Piano/Music	566 Students	ChatGPT	Experimental (TTCT)	Significant increase in creative skills (Torrance Tests)
Grájeda et al. [19]	Visual Arts	180 Students	ChatGPT	Pilot Survey (30-Q)	Emotional reaction (joy/surprise); Ethical concerns
Beniwal and Das [3]	Design (Basic)	32 Faculty/Students	AI (General)	Online Survey	Balanced approach; Ethical frameworks; Industry alignment
Kollia et al. [32]	Engineering/Creative	10 Students	ChatGPT 3.5	Experimental + Quest	Enhanced fluency/flexibility; No gain in text originality
Hatmanto et al. [20]	Education (Philippines)	10 Senior Educators	ChatGPT	Semi-structured Interviews	Curriculum mapping; Strategic integration requirements
Zhao [45]	Music Teaching	Data Analysis	ChatGPT	Quantitative (MATLAB)	Optimization of teaching personalization
Fleischmann [17]	Design (Postgrad)	33 Students	GenAI	Interviews	Career anxiety (FOMO); Industry readiness
Li and Tu [33]	Expressive Arts	10 Students	Text-to-Image	Qual (Journals) + Quant	Therapeutic benefits: Self-awareness and confidence
Luo and Tahir [35]	STEAM Education	19 Educators	ChatGPT	Qual (Interviews/Records)	Lesson planning efficiency; Prompt framework development
Zhu et al. [47]	Digital Illustration	80 Students	GenAI + A/R/Tography	Quasi-experimental	Heritage pattern design; Cross-cultural dialogue
Jiang [28]	Music Philosophy	N/A (Theoretical)	GenAI	Empirical Instances Study	Philosophical/Ethical questions in music AI
Bell [2]	Music Production	N/A (App Analysis)	Google Music Tools	Empirical Instances Study	Critique of "black box" magic vs. algorithmic logic

Table 2 (continued)

Author (s) & Year	Context/discipline	Participants	GenAI Tool (s)	Methodology	Key focus/outcome
Xie and Hong [44]	Art & Design	Undergrads	NLP/Deep Learning	Experimental + Interviews	Educational content generation mechanism
Knochel [31]	Postdigital Art	N/A (Theory)	Photoshop/ Midjourney	Critical Theory	Human-AI assemblages; Data ownership ethics
Tsidylo and Sena [41]	Design Training	39 Students	DALL-E 2/ Midjourney	Questionnaire	Methodological innovation vs. Lack of fundamental knowledge
Huh et al. [25]	Architecture	42 Students	Image GenAI	Online Survey	Acceptance by design stage (Concept vs. Final)
He and Ren [21]	Music (Pre-service)	301 Teachers	GenAI	Questionnaire (UTAUT2)	Factors of acceptance: Performance Expectancy, Social Influence
Liu and Liao [34]	Flute Learning	~30 Students	IBM Watson Beat	Questionnaire	Learning strategies; Deep vs. Surface motive indices
Dehouche and Dehouche [13]	Art History	N/A (Data)	72,980 Prompts	Case Study	Prompt engineering as art history pedagogy; Copyright

intent. Particular attention was paid to tensions repeatedly reported between enhanced creative ideation and limitations in technical execution, authorship, or material specificity. Through constant comparison, these recurring tensions were synthesized into broader conceptual patterns.

Third, the emergent themes were interpreted in relation to established theories of creativity, learning, and scaffolding. The concept of the "ideation–execution gap" emerged inductively from this process as a cross-disciplinary pattern describing the asymmetry between GenAI's effectiveness in early-stage creative exploration and its reduced reliability in supporting high-fidelity artistic execution. This interpretive step allowed the synthesis to move beyond thematic description toward conceptual integration, linking empirical findings to pedagogical and theoretical implications.

To enhance transparency and analytical rigor, theme development was iteratively reviewed against the full corpus to ensure that identified patterns were consistently supported across multiple studies rather than driven by isolated cases. Discrepancies or discipline-specific divergences were retained and discussed rather than subsumed into a single unified model, reflecting the diversity of practices within higher art education.

Results

The systematic review of the database, comprising 27 empirical and theoretical studies published primarily between 2023 and 2025, elucidates a higher art education landscape currently undergoing profound epistemological and pedagogical shifts. The integration of Generative Artificial Intelligence (GenAI) has transitioned from speculative adoption to a phase of rigorous empirical interrogation. This body of work is characterized by methodological pluralism, ranging from large-scale quantitative inquiries to intimate, bi-ethnographic qualitative studies. The aggregated data suggests that GenAI functions not merely as an instrumental utility but as a disruptive agent, necessitating a re-evaluation of creativity, authorship, and skill acquisition across visual arts, music, design, and interdisciplinary STEAM education.

The following analysis synthesizes findings into six thematic clusters: (1) the transformation of creative workflows, specifically the dichotomy between ideational fluency and technical execution; (2) discipline-specific integration trajectories; (3) cognitive and affective learning outcomes; (4) emergent pedagogical frameworks; (5) stakeholder perceptions and institutional readiness; and (6) the ethical and technical complexities of algorithmic creativity.

Educational Contexts and Participant Demographics

The articles reflects a substantial broadening of the research scope, both geographically and methodologically. The included studies span diverse educational settings, from music conservatories in China to multimedia design studios in Jordan and teacher training programs in the Philippines. Participant demographics have evolved from small-scale pilot groups to robust cohorts, exemplified by Wang's [42] experimental study of 566 music students and He and Ren's [21] survey of 301 pre-service teachers. Simultaneously, qualitative depth is preserved through micro-studies, such as Ansone et al.'s [1] observational research and Ma et al.'s [36] bi-ethnographic inquiry. This methodological heterogeneity indicates a maturation of the field towards evidence-based validation of AI interventions.

The Transformation of Creative Ideation and Production Workflows

A predominant theme emerging from the literature is the bifurcation of the creative process into distinct phases of "ideation" and "execution," with GenAI acting as a potent accelerator for the former while introducing friction in the latter.

The "Ideation Engine": Enhancing Fluency and Divergence

Across multiple disciplines, GenAI has been validated as a superior instrument for the ideation phase of artistic production. In the visual arts, Ansone et al. [1] observed that students utilizing DALL-E perceived the tool as an "inspirational and conceptual mentor," capable of generating visual variations at a velocity unattainable through manual means. This finding aligns with Kollia et al. [32], whose experimental data indicated that ChatGPT significantly enhanced "fluency" and "flexibility" in divergent thinking tasks among engineering students, acting as a catalyst for cognitive expansion. Similarly, in multimedia design, Khasawneh et al. [30] demonstrated that the use of Midjourney facilitated rapid prototyping of character morphologies, thereby promoting a culture of "innovative thinking" and workflow optimization.

The "Execution Gap": Precision and Authorship

Conversely, the data reveals a critical "execution gap" where the utility of GenAI diminishes as tasks require greater nuance and specific artistic intent. Ansone et al. [1] reported that students experienced frustration with the tools' limitations in executing "nuanced artistic tasks," citing a lack of spatial control and color consistency. This limitation is corroborated by Huh et al. [25] in architectural education, where student acceptance of GenAI declined significantly from the conceptual design phase to the final presentation phase. Participants expressed concern that reliance on AI for detailed development compromised "originality and critical judgment," essential components of professional identity. These findings suggest a U-shaped utility curve for GenAI, where its value is maximized at the initial brainstorming stage but requires substantial human intervention for final refinement.

Cognitive and Psychological Impacts: Creativity, Motivation, and Self-Efficacy

The review highlights significant cognitive and affective implications of GenAI integration, moving beyond anecdotal evidence to measure specific psychological constructs.

Quantitative Measures of Creative Enhancement

Wang's [42] study provides robust empirical evidence regarding creativity. Utilizing the Torrance Tests of Creative Thinking (TTCT), the study found a statistically significant increase in creative skills within the experimental group using ChatGPT for piano instruction compared to the control group. The conversational interaction with AI appeared to stimulate divergent thinking patterns, facilitating a broader exploration of musical theory. However, Kollia et al. [32] offer a necessary counterpoint, noting that while AI enhanced the quantity and variety of ideas ("fluency" and "flexibility"), it did not yield a statistically significant improvement in "originality" for creative text production. This discrepancy underscores the discipline-dependent nature of AI-mediated creativity.

Affective Outcomes and the Reduction of Anxiety

In the affective domain, GenAI appears to lower the "affective filter" associated with artistic performance. Wang and YeaJin (2024) reported that ChatGPT-4 fostered a "perception of tranquility and ease" among music students, acting as a non-judgmental tutor that reduced performance anxiety. Similarly, Chen et al. [8] found that GenAI usage in a design history course positively correlated with student motivation and self-efficacy. Furthermore, Li and Tu [33] identified therapeutic benefits in Expressive Arts Counseling Courses, where text-to-image tools enhanced participants' self-awareness and confidence by allowing them to visualize internal states that were difficult to articulate verbally.

Discipline-Specific Integration: A Comparative Analysis

The integration of GenAI manifests distinctly across different artistic sub-disciplines.

Music Education: This field exhibits the most mature empirical validation. Large-scale studies utilizing standardized instruments (e.g., TTCT, UTAUT2) demonstrate that GenAI tools like IBM Watson Beat effectively support theoretical acquisition and compositional complexity. He and

Ren [21] identified "Performance Expectancy" as a primary driver for AI adoption among pre-service music teachers.

Visual Arts and Design: In contrast, these disciplines are characterized by a tension between efficiency and authenticity. While Khasawneh et al. [30] highlight efficiency gains, Tsidylo and Sena [41] warn of a potential "de-skilling" effect, observing that students using DALL-E 2 displayed a "lack of fundamental knowledge" regarding both neural networks and visual arts principles.

STEAM and Interdisciplinary Education: Here, GenAI functions primarily as an integrative scaffold. Luo and Tahir [35] demonstrated ChatGPT's efficacy as a co-planner for STEAM curricula, synthesizing interdisciplinary resources more effectively than traditional methods.

Pedagogical Innovations and Frameworks

Educators are increasingly moving beyond ad-hoc tool adoption to develop structured pedagogical frameworks.

A/R/Tography and Cultural Heritage: Zhu et al. [47] integrated GenAI with A/R/Tography (Artist/Researcher/Teacher) in a digital illustration curriculum. The study found that while AI facilitates cross-cultural dialogue through the reconstruction of heritage patterns, "human intervention remains essential for cultural accuracy," positioning the student as a critical curator.

Prompt Engineering as Academic Literacy: Dehouche and Dehouche [13] argue that prompt engineering constitutes a new form of art history education, requiring a deep vocabulary of aesthetics and style to be effective. This aligns with Luo and Tahir's [35] development of a "prompt framework" for lesson planning, emphasizing that the quality of output is contingent upon the user's domain knowledge.

Challenges: Ethics, Bias, and the "Black Box"

Despite the benefits, significant challenges persist. Bell [2] critiques the "magic" of AI music systems, arguing that their "black box" nature obscures algorithmic logic, potentially reducing students to passive operators. Ethical concerns regarding data provenance are also prominent, Knochel [31] emphasizes the moral implications of using tools trained on contested datasets, calling for a curriculum that addresses "training data ownership". Additionally, the "readiness gap" identified by 9, 10 reveals that specialized art colleges exhibit greater cutural resistance to AI compared to comprehensive universities, highlighting a divide in institutional preparedness.

Discussion

Interpretation of Key Findings

The synthesis of the 27 reviewed studies confirms that higher art education is undergoing a pronounced "generative turn," in which generative AI is transitioning from an experimental novelty to an increasingly central pedagogical agent. Across disciplines, including visual arts, design, music, and interdisciplinary creative practices. GenAI tools are most consistently positioned as supports for early-stage creative exploration rather than as substitutes for artistic expertise.

A central and recurring pattern identified in the review is the ideation–execution gap. While GenAI tools function as powerful "ideation engines" that democratize access to complex conceptual exploration and accelerate divergent idea generation [1], they frequently fall short during the execution phase, where technical nuance, material sensitivity, and contextual judgment are required. This gap suggests that current generative systems are better understood as collaborative scaffolds that augment, rather than replace, human creative agency.

From a creativity theory perspective, this phenomenon aligns with the distinction between divergent and convergent thinking. The reviewed studies indicate that GenAI excels in supporting divergent processes—such as brainstorming, stylistic variation, and conceptual recombination—while convergent processes involving refinement, evaluation, and technical realization remain largely human-dependent. From a learning theory perspective, this dynamic can be further interpreted through cognitive load and situated learning frameworks. Generative AI appears to reduce cognitive and affective load during ideation, particularly for novices, but execution requires embodied engagement, iterative practice, and domain-specific knowledge situated within cultural and material contexts. These demands help explain the persistence of the ideation–execution gap across educational settings.

Psychologically, this duality produces differentiated effects across learner populations. While several studies report reduced anxiety and increased creative confidence among beginners (Wang and YeaJin, 2024), others document heightened concern among advanced students and emerging professionals regarding authorship, originality, and long-term professional relevance in AI-augmented creative fields [17].

Implications for Teaching Practice

The findings of this review suggest that effective GenAI integration in higher art education requires a shift from tool adoption toward critical pedagogical integration.

Prompt Engineering as Core Creative Literacy

Rather than being treated as a procedural or technical skill, prompt engineering should be framed as a form of conceptual and critical literacy grounded in art history, aesthetics, and disciplinary conventions. The reviewed studies suggest that meaningful prompting depends on students' ability to articulate artistic intent and critically evaluate outputs, rather than on surface-level command usage alone [13].

The Human-in-the-Loop Paradigm

Pedagogical models should emphasize students' roles as curators, editors, and reflective agents in AI-assisted creation. Frameworks such as A/R/Tography [47] underscore the importance of human judgment in mediating AI outputs, reinforcing that artistic meaning, cultural context, and ethical responsibility remain irreducibly human concerns.

Technical Access and Pedagogical Affordances

Across the reviewed studies, a clear pattern emerges regarding the technical implementation of GenAI tools and their corresponding pedagogical affordances. The overwhelming majority of empirical studies relied on web-based or platform-mediated interfaces, including conversational web interfaces (e.g., ChatGPT), Software-as-a-Service image generators (e.g., DALL·E), and community-based platforms such as Midjourney via Discord [1, 30],Wang and YeaJin, 2024). These deployment modes prioritize ease of access, low technical barriers, and rapid classroom integration, making them particularly suitable for large cohorts, introductory courses, and ideation-focused learning activities.

Within these interface-driven contexts, GenAI tools consistently supported conceptual exploration, motivation, and perceived ease of use, especially in early or theory-oriented stages of learning [8, 42]. However, multiple studies also report limitations when such tools are applied to tasks requiring fine-grained technical control or stylistic nuance, reinforcing the persistence of the ideation–execution gap [1, 41]. These findings suggest that interface-based tools tend to foreground generative breadth over depth, shaping how students engage with creative processes.

By contrast, a smaller but pedagogically significant subset of studies employed API-based, integrated, or locally installed systems, often within more specialized or advanced educational settings. For example, Chiu and Hwang [11] and Xie and Hong [44] incorporated API-based or custom mechanisms to embed GenAI within structured learning environments, enabling closer alignment with critical thinking objectives and curricular scaffolding. Similarly, studies involving locally hosted or open-source models (e.g., IBM Watson Beat, Stable Diffusion, or open prompt datasets) emphasized deeper engagement with learning strategies, model logic, or historical and ethical dimensions of AI-assisted creativity [13, 31, 34].

Importantly, these more technically demanding implementations were not primarily framed as engineering exercises but as pedagogical interventions that afford greater transparency, reflective engagement, and awareness of algorithmic agency. However, they also introduce higher demands on institutional infrastructure, instructor expertise, and student technical literacy—factors that may limit scalability and exacerbate inequities across educational contexts [9, 10].

Taken together, the reviewed literature suggests that technical deployment mode is not a neutral choice but a pedagogical one. Web-based interfaces tend to support accessibility, motivation, and ideational fluency, while API-based or local implementations afford deeper critical engagement, executional control, and ethical reflection. Effective curriculum design in higher art education therefore requires aligning GenAI deployment strategies with learning objectives, student preparedness, and institutional capacity, rather than assuming a one-size-fits-all model of AI integration.

Ethical Considerations for Generative AI in Higher Art Education

Ethical concerns related to generative AI were present across the reviewed literature but often addressed in fragmented or case-specific ways. Synthesizing these discussions reveals three interrelated ethical dimensions that are particularly salient for educational contexts: authorship and intellectual property, bias and cultural representation, and data responsibility and algorithmic transparency. Together, these dimensions shape how GenAI tools are perceived, adopted, and pedagogically governed in higher art education.

Authorship and Intellectual Property

Questions surrounding copyright, originality, and authorship remain unresolved in AI-assisted creative practice and are especially consequential in educational settings, where assessment, attribution, and learning outcomes are at stake. Several studies highlight student uncertainty regarding ownership of AI-generated outputs, particularly when training data and generative processes remain opaque [19, 31]. Dehouche and Dehouche [13] further demonstrate that prompt engineering itself can constitute an intellectual and historical act, complicating simplistic distinctions between "human-made" and "machine-made" work. From a pedagogical perspective, this suggests a need to shift assessment frameworks away from product-centric originality toward process-oriented criteria that recognize conceptual intent, critical intervention, and reflective authorship within human–AI assemblages.

Bias, Representation, and Cultural Integrity

A second ethical concern relates to the reproduction of cultural bias and normative aesthetics embedded within generative models. Multiple studies caution that GenAI systems often privilege dominant visual styles, Western art historical canons, or commercially popular aesthetics, potentially marginalizing minority cultural expressions and alternative epistemologies [2, 47]. In higher art education, such tendencies risk reinforcing homogenization rather than fostering pluralistic creativity. Pedagogical interventions that foreground cultural critique, dataset awareness, and cross-cultural dialogue—such as heritage-based design projects or comparative output analysis—emerge as important counterbalances to these risks. Ethical AI literacy, therefore, must include not only technical awareness but also critical engagement with questions of representation, power, and cultural authorship.

Data Responsibility and Algorithmic Transparency

A third ethical dimension concerns data provenance, model opacity, and students' limited understanding of how generative systems operate. Several reviewed studies identify the "black box" nature of GenAI as a barrier to meaningful learning, particularly when tools are treated as magical or autonomous creative agents rather than socio-technical systems shaped by human design choices [2, 28]. From an educational standpoint, algorithmic transparency is less about technical mastery and more about cultivating informed skepticism and responsible use. Approaches that expose students to simplified model logic, training data debates, or comparative system behaviors help reposition GenAI as an object of inquiry rather than an unquestioned authority.

Taken together, these ethical considerations suggest that responsible GenAI integration in higher art education cannot be addressed through isolated policy statements or technical safeguards alone. Instead, ethics should be embedded as an ongoing pedagogical concern—interwoven with studio critique, reflective writing, and curricular design. Such an approach aligns ethical awareness with creative practice, ensuring that students develop not only technical fluency but also critical responsibility as emerging artists, designers, and educators in AI-augmented cultural landscapes.

Conclusion

This systematic review demonstrates that Generative AI is not merely an auxiliary technology but a transformative force reshaping both the epistemology and pedagogy of higher art education. Synthesizing 27 recent empirical studies, the review clarifies how GenAI consistently enhances ideational fluency, lowers affective barriers, and broadens access to creative exploration, while simultaneously exposing a persistent ideation–execution gap in which technical refinement, embodied skill, and contextual sensitivity remain largely human-dependent.

The primary contribution of this review lies in moving beyond tool-centric discussions toward an integrated pedagogical and theoretical account of GenAI in art education. By systematically linking learning outcomes to creativity theory, learning theory, and modes of technical implementation, this study shows that GenAI's educational value depends less on the tools themselves than on how they are pedagogically framed, technically deployed, and ethically governed. In particular, the distinction between interface-based and more transparent or locally implemented systems reveals that technical access is a pedagogical choice with direct implications for depth of learning, critical engagement, and equity.

The findings underscore the necessity of Human-in-the-Loop pedagogies, in which students are trained as reflective curators, critics, and decision-makers rather than passive operators of algorithmic systems. Embedding critical AI literacy—including prompt design as conceptual practice, ethical awareness of authorship and bias, and foundational understanding of algorithmic processes—emerges as essential for preserving artistic agency in AI-augmented creative environments.

Future research should extend beyond short-term classroom interventions to examine longitudinal effects on skill development, creative identity, and professional trajectories. Comparative studies across GenAI architectures and institutional contexts are also needed to address emerging inequities in access and expertise. Ultimately, the question for higher art education is no longer whether GenAI should be used, but how it can be integrated in ways that sustain technical mastery, cultural integrity, and deep creative learning in the long term.

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Data Availability The data supporting the findings of this study are available from the corresponding author upon reasonable request. No publicly archived datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The author declares no competing interests.

Research Involving Human and/or Animals. This study did not involve human participants or animals.

Informed Consent Not applicable, as this study involved no human participants requiring informed consent.

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